CSA5108 Cryptography and Network security

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Exp 8 : Simple Substitution Cipher
Code:

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main(void) {
   char key[100], text[1000], cipher[27];
   int used[26] = {0}, k = 0;

   printf("Enter keyword: ");
   scanf("%99s", key);

   for (int i = 0; key[i]; i++) {
      char c = toupper((unsigned char)key[i]);
      if (c >= 'A' && c <= 'Z' && !used[c - 'A']) {
         cipher[k++] = c;
         used[c - 'A'] = 1;
      }
   }

   for (char c = 'A'; c <= 'Z'; c++)
      if (!used[c - 'A'])
         cipher[k++] = c;

   cipher[26] = '\0';

   printf("Cipher alphabet: %s\n", cipher);

   getchar();
   printf("Enter plaintext: ");
   fgets(text, sizeof(text), stdin);
```

```
    for (int i = 0; text[i]; i++) {
        if (isupper((unsigned char)text[i]))
            text[i] = cipher[text[i] - 'A'];
        else if (islower((unsigned char)text[i]))
            text[i] = tolower((unsigned char)cipher[text[i] - 'a']);
    }

    printf("Ciphertext: %s", text);
    return 0;

}
```

